

Econ 30065: Microeconomics

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Choices under uncertainty

Risk aversion

- A person is ***risk averse*** if, in comparing a riskless bundle to a risky bundle with the same expected value, he prefers the riskless bundle
 - Concave utility function
 - Degree of risk aversion may vary from person to person, or with age for the same person
 - The greater the concavity, the greater the risk aversion
- For a ***risk-averse*** person, the certainty equivalent of a risky bundle is always less than expected value
 - Providing the same expected value with no risk would make the individual better off
- For a ***risk-averse*** person, the risk premium is positive

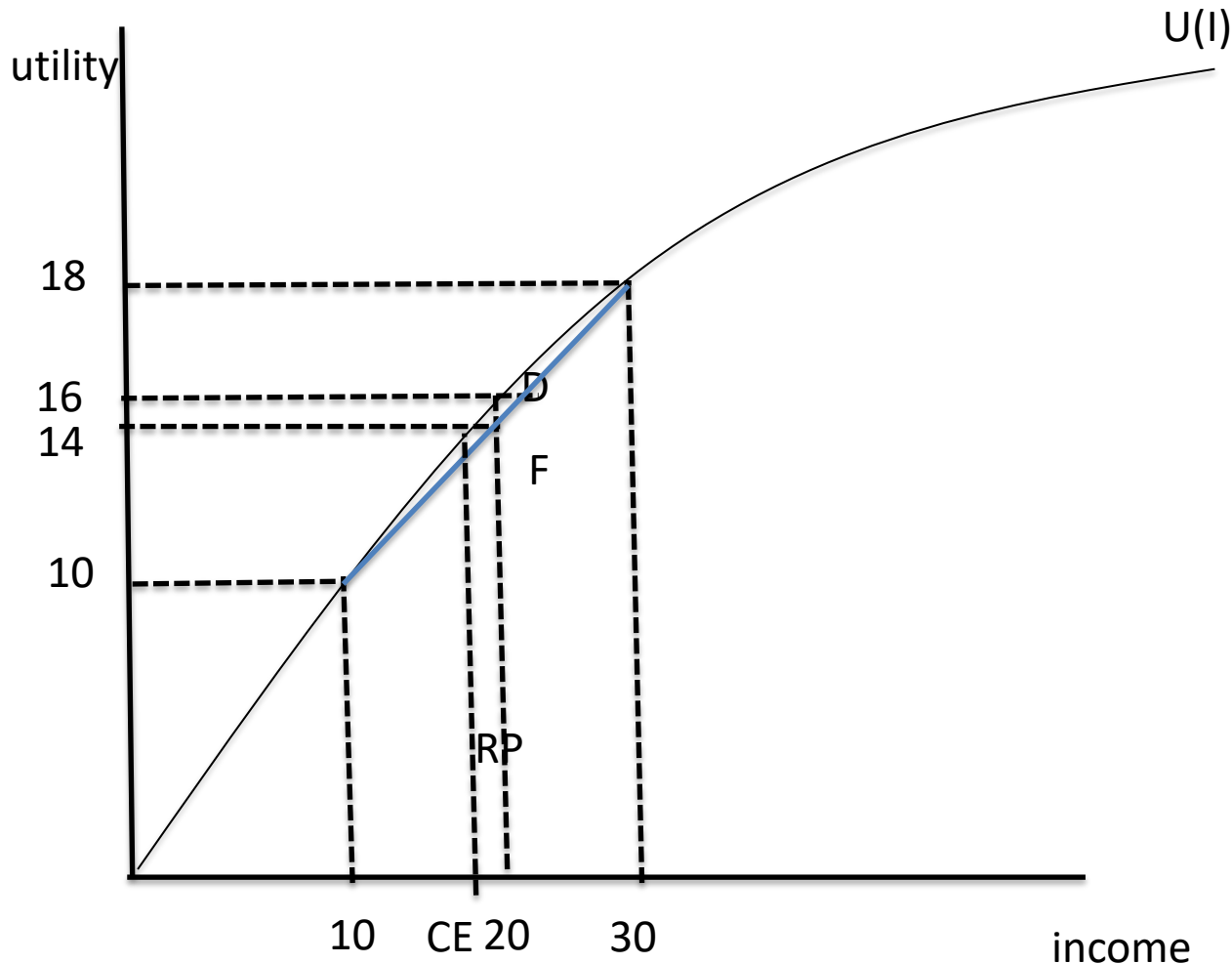
Risk aversion example

- A person can have a \$20,000 job with probability 1 and receives a utility level of 16
- The person could have a job with a 0.5 chance of earning \$30,000 and a 0.5 chance of earning \$10,000.
- Utility at \$30,000 is 18
- Utility at \$10,000 is 10

Risk aversion example

- Expected value of Risky Job
 $EV = 0.5 \times 30000 + 0.5 \times 10000 = 20000$
- Expected Utility of Risky Job
 $E(U) = 0.5(18) + 0.5 \times 10 = 14$
- Expected value from both jobs is the same – risk averse may choose current job
- Expected utility is greater for certain job
 - Would keep certain job

Risk averse utility function

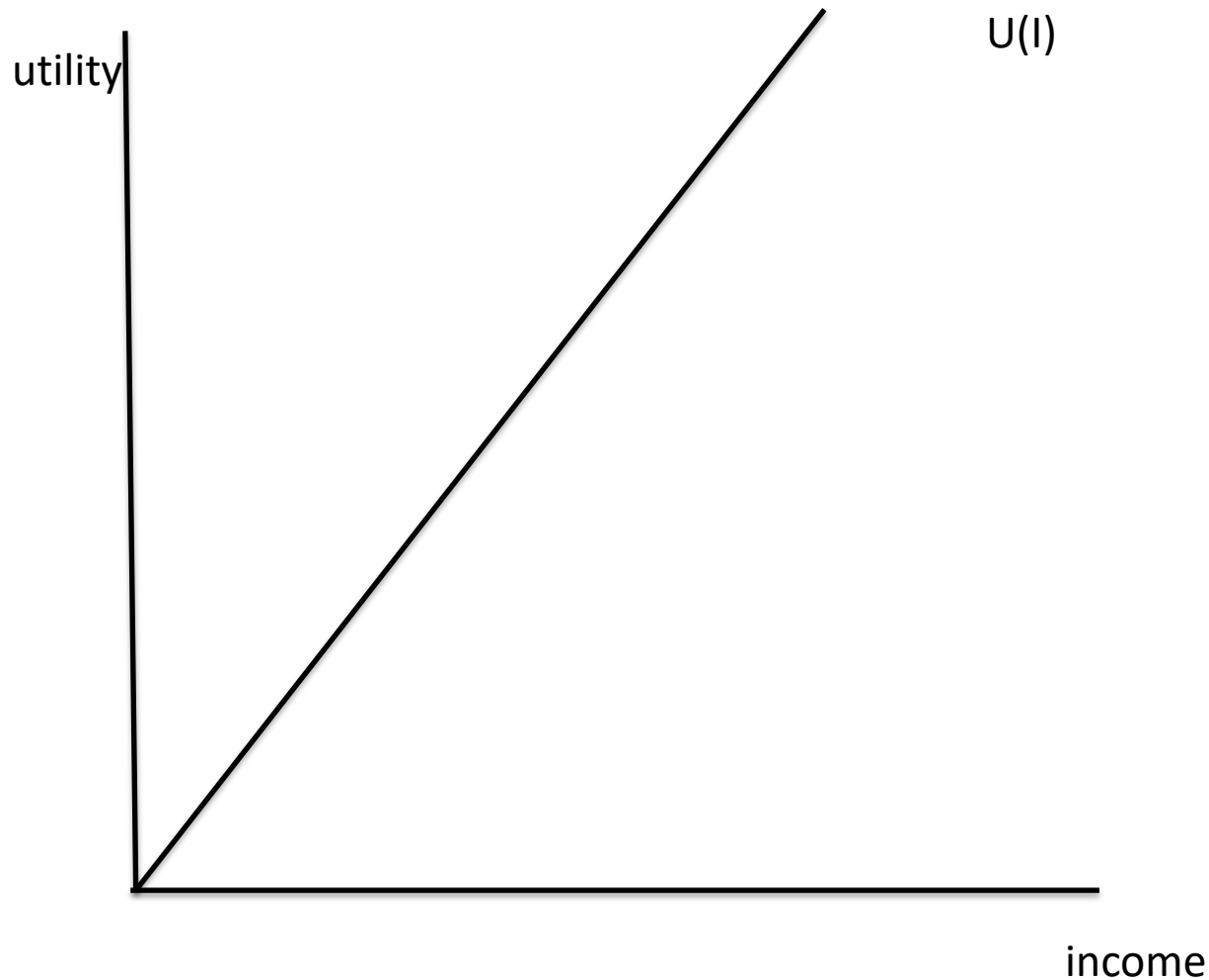


- Risky job has expected value of \$20,000 with expected utility of 14
 - Point F
- Certain job has expected value of \$20,000 with utility of 16
 - Point D

Risk neutrality

- A person is ***risk neutral*** if, in comparing a riskless bundle to a risky bundle with the same expected value, he is indifferent between the two
 - Linear utility function
- For a ***risk-neutral*** person, the certainty equivalent of a risky bundle is always equal to the expected value
 - Providing the same expected value with no risk would make the individual indifferent
- For a ***risk-neutral*** person, the risk premium is zero

Risk neutrality utility function



Risk loving

- A person is ***risk lover*** if, in comparing a riskless bundle to a risky bundle with the same expected value, he prefers the risky bundle
 - Convex utility function
 - The greater the convexity, the greater the risk loving
- For a ***risk-loving*** person, the certainty equivalent of a risky bundle is always more than expected value
 - Providing the same expected value with no risk would make the individual worse off
- For a ***risk-loving*** person, the risk premium is negative

Risk loving example

A person can have a \$20,000 job with probability 1 and receive a utility level of 8. The person could have a job with a 0.5 chance of earning \$30,000 and a 0.5 chance of earning \$10,000. Utility at \$30,000 is 18 and Utility at \$10,000 is 3.

Risk loving example

- Expected value of Risky Job

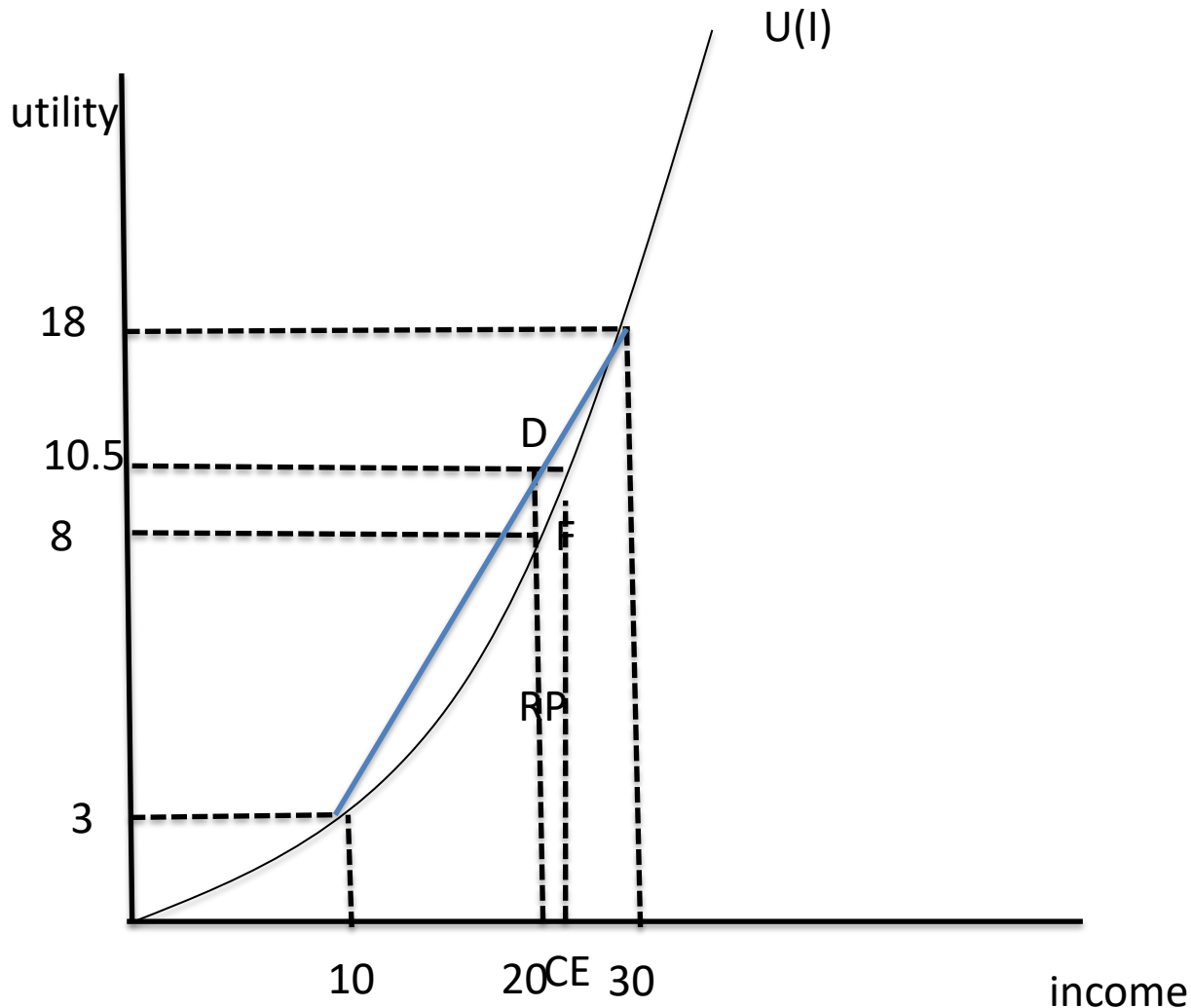
$$E(I)=0.5 \times 30000 + 0.5 \times 10000 = 20000$$

- Expected Utility of Risky Job

$$E(U)=0.5(18) + 0.5 \times 3 = 10.5$$

- Expected income from both jobs is the same
- Expected utility is greater for uncertain job
 - Would take uncertain job

Risk loving utility function



- Risky job has expected income of \$20,000 with expected utility of 10.5
 - Point D
- Certain job has expected income of \$20,000 with utility of 8
 - Point F

Example 1

➤ Consider the following lottery:

Probability	Payoff (x)
1/3	\$9
1/3	\$36
1/3	\$81

- Silvio's utility function is given by $u(x) = \sqrt{x}$
- Calculate the expected value of the lottery and Silvio's expected utility from participating in this lottery.
 - Would he prefer the lottery or to receive its expected value?
 - Calculate the certainty equivalent of the lottery. What's Silvio's risk premium?

Example 1 – solution:

- $EV = 1/3 \times 9 + 1/3 \times 36 + 1/3 \times 81 = 42$
- $EU = 1/3 \times u(9) + 1/3 \times u(36) + 1/3 \times u(81) = 1 + 2 + 3 = 6$
- Since $u(\cdot)$ is concave, Silvio is risk averse. He will definitely prefer the safe option of taking 42 for sure.
- To find CE, solve $U(CE) = EU \rightarrow CE = 36$
- Risk premium $= EV - CE = 42 - 36 = 6 > 0$ since he's risk averse

Example 2

➤ Consider the following lottery:

Probability	Payoff (x)
1/2	\$2
1/4	\$4
1/4	\$8

- Silvia's utility function is given by $u(x) = x^2$
- Calculate the expected value of the lottery and Silvia's expected utility from participating in this lottery.
 - Would she prefer the lottery or to receive its expected value?
 - Calculate the certainty equivalent of the lottery. What's Silvia's risk premium?

Example 2 – solution:

- $EV = 1/2 \times 2 + 1/4 \times 4 + 1/4 \times 8 = 4$
- $EU = 1/2 \times u(2) + 1/4 \times u(4) + 1/4 \times u(8) = 22$
- Since $u(\cdot)$ is convex, Silvia is risk loving. She will definitely prefer the lottery to having 4 for sure.
- To find CE, solve $U(CE) = EU \rightarrow$
 $CE = \sqrt{22} \approx 4.7$
- Risk premium $= EV - CE = 4 - 4.7 = -0.7 < 0$ since she's risk-loving